

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. STA/STB-503: Programming with C

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.(a) Evaluate the following C expressions (assume `int` arithmetic):(i) $15 - 10 / 3$ (ii) $10 / 4.0 - 2 * 5$ (iii) $5 / 3 - 2 - 5 * 2$ (iv) $-3 \% -8 / 2 + 7$

[7]

(b) With the help of examples, explain numeric, character, real, alphanumeric, and string constants in C. [7]

Question 2.(a) With the help of a program, explain the use of increment (`++`) and decrement (`--`) operators. Write a C program to find the maximum of three numbers. [7]

(b) Write the basic data types used in C. Also explain enumerated data types with an example. [7]

Question 3.(a) What is the purpose of the `for` statement? How does it differ from the `while` statement? Explain with the help of a program. [7](b) Write a C program to perform the *t*-test for testing equality of two means. [7]

Question 4.(a) Explain the use of `break`, `continue`, and `goto` statements with the help of a program. [7]

- (b) With the help of a program, explain the use of the `switch` statement in C. [7]
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Question 5.

- (a) Define one-, two-, and multi-dimensional arrays in C. Write a program for multiplication of two matrices. [7]
- (b) Write a C program for fitting a straight line $Y = a + bX$ by the method of least squares. [7]
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Question 6.

- (a) With the help of a program, explain automatic, static, and external variables in C. [7]
- (b) Explain functions with arguments and return values. Write a C program to find the median of n given numbers using a function. [7]
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Question 7.

- (a) What do you mean by recursion? Write a C program for the generation of the Fibonacci series using recursion. [7]
- (b) Write a C program for calculating the Karl Pearson correlation coefficient using a function. [7]
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Question 8.

- (a) Define structure in C. Write a C program to show the use of structures. [7]
- (b) Describe unions in C. Write the similarities and differences between structures and unions, with a program demonstrating the use of a union. [7]
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